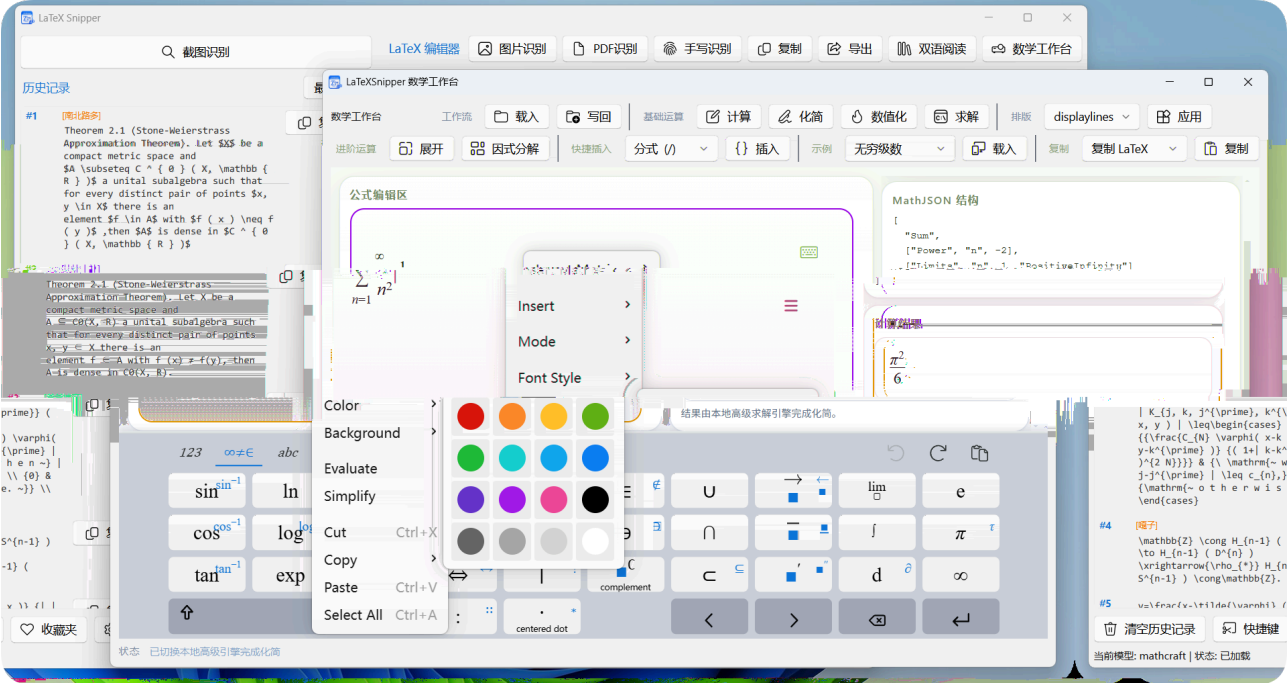


LaTeXSnipper



• LaTeXSnipper

• MathCraft

· LaTeXSnippet



python3 -m venv

python3-venv



80%

→ → / →

Windows %USERPROFILE%\latexsnippet\logs\ %LOCALAPPDATA%\LaTeXSnippet\logs\

Linux ~/.latexsnippet/logs/

macOS ~/.latexsnippet/logs/

crash-native.log



Ctrl+F

PDF

/

/

+

PDF

PDF

Poppler Fitz

Argos Translate Azure Translator Google Cloud Translation

DeepL API Free

GPU
GPU

MATHCRAFT_FORCE_ORT_CPU=1

/

invalid protobuf failed to load model sha256 mismatch

%APPDATA%\MathCraft\models\<model_id>\

~/.mathcraft\models/<model_id>/

python -m mathcraft_ocr models check

ONNX

MATHCRAFT_FORCE_ORT_CPU=1



```
def _validate_config(model_name, base_url):
```

→

A

B

C

```
ollama run <model_name>
```

Ollama

Base URL

```
http://localhost:11434/v1
https:// http://
```

```
ollama serve
```



```
http://127.0.0.1:11434/api/tags
```

/

```
ollama list
```

```
/v1/models
```

API

API Key





MinerU

`/health` `/file_parse`

`http://127.0.0.1:8000/health`

OpenAI-compatible

Ollama

`/v1/models` `/api/tags` `/v1/models`

!

Ollama

MinerU

OpenAI-compatible

Python

Windows
Linux/macOS
/

创建虚拟环境

激活

激活

安装依赖

PyInstaller

```
git clone
```

requirements.txt requirements-build.txt

Linux/macOS

```
tools/deps/python311
```

!

/usr/lib/latexsnipper .app

~/.latexsnipper/deps/python311

pyvenv.cfg

.deb python3 python3-venv .deb

python3

或使用

官方

安装包

pip

pywin32 / PyQt6

```
pip install -r requirements.txt
```

```
ensurepip pip setuptools wheel
```

python-3.11.0-amd64.exe

pywin32

PyQt6

pywin32

Program Files /

C:\Program Files\

%APPDATA% ~/.latexsnipper/

%LOCALAPPDATA%

/

%APPDATA%

MATHCRAFT_HOME

C:\mathcraft_data

Windows

0.0.0.0 127.0.0.1

api.github.com

%USERPROFILE%\.latexsnipper/logs/
/

GitHub Releases

PDF

PDF

/

Linux/macOS

Pandoc

/

/

amsmath amssymb mathtools geometry

*

2x

2*x

/

Pandoc

.docx

.epub

ctex

PDF

PDF

Linux / macOS

pynput

Ctrl+字母 Ctrl+Shift+字母

Ctrl+F

```
grim maim gnome-screenshot
```

screenshot

/

~/[.latexsnipper/deps/python311](#)

.deb

.dmg

.app.zip

!

Ctrl+字母 Ctrl+Shift+字母

Linux: Wayland

grim slurp

gnome-screenshot

! Linux

```
grim maim gnome-screenshot
```

Linux: Wayland

EGL not available No physical devices GL0zone not found Failed to get system
egl display

/dev/dri/renderD*

强制启用

图形兜底

禁用

图形兜底，尝试原生

路径

macOS:

xattr -cr /Applications/LaTeXSnippet.app

macOS:

macOS:

macOS: pip pip

pip pip setuptools wheel

~/.latexsnipper/deps/python311 venv ensurepip pip

brew install python

配置文件 日志：

依赖环境：

模型缓存（ ）：

Python

where python

which python

/

Bug

X

“ ”“ ”

issue

1.

%USERPROFILE%\latexsnipper\logs\ %LOCALAPPDATA%\LaTeXSnipper\logs\

logs

2.

crash-native.log

3.

4.

5.

6.

GitHub Issues
GitHub Discussions

README Quick Start



Windows `python -m venv .venv pip install -r requirements.txt python src/main.py`
Linux `python3 -m venv .venv pip install -r requirements-linux.txt python src/main.py`
macOS `python3 -m venv .venv pip install -r requirements-macos.txt python src/main.py`
`requirements-linux.txt requirements-macos.txt -r requirements.txt`
`pynput`

`python311/ tools/deps/python311/`

`python311/`

`tools/deps/python311/`
`ruff pytest pyright`

`requirements*.txt`

`tools/deps/python311`

`~/.latexsnipper/`

`deps/python311`

Windows



`python311/`
`tools/deps/python311/`
`~/.latexsnipper/deps/python311`

`docs/developer_code_standards.md`



`ruff pytest pyright requirements*.txt`

pyrightconfig.json

tools/deps/python311

python311/

	tools/deps/python311	python311
---	----------------------	-----------

MathCraft OCR

ONNX

检查模型缓存状态

手动下载模型

识别图片中的公式

LaTeXSnipper

mathcraft-ocr	0.2.0	v1.0.0	4
ONNX			
mathcraft-formula-det			
mathcraft-formula-rec			
mathcraft-text-det			
mathcraft-text-rec			
!			

Profiles

formula

text

mixed

Abstract Algebra (p.18)

CHAPTER 1. PRELIMINARIES

0

PROOF. (1) If $A \cup B = \emptyset$, then the theorem follows immediately since both A and B are empty sets. Otherwise, we must show that $(A \cup B) \subseteq A' \cap B'$ and $(A \cup B) \supseteq A' \cap B'$. Let $x \in A \cup B$. Then $x \in A$ or $x \in B$. So $x \in A'$ and $x \in B'$ by definition of the union of sets. By the definition of the complement, $x \in A' \cap B'$ and we have $(A \cup B) \subseteq A' \cap B'$.

To show the reverse inclusion, suppose that $x \in A' \cap B'$. Then $x \in A'$ and $x \in B'$, and so $x \notin A$ and $x \notin B$. Hence $x \in A \cup B$. The proof of (2) is left as an exercise.

Example 1.4 Other relations between sets often hold true. For example,

$$(A \setminus B) \cap (B \setminus A) = \emptyset$$

To see that this is true, observe that

$$\begin{aligned}(A \setminus B) \cap (B \setminus A) &= (A \cap B') \cap (B \cap A') \\ &= A \cap A' \cap B \cap B' \\ &= \emptyset.\end{aligned}$$

Cartesian Products and Mappings

Given sets A and B , we define a new set $A \times B$ called the **Cartesian product** of A and B consisting of ordered pairs. That is,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Example 1.5 If $A = \{x, y\}$ and $B = \{1, 2, 3\}$, then $A \times B$ is the set

$$\{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$$

and

$$A \times A = \{(x, x), (x, y), (y, x), (y, y)\}.$$

We define the **Cartesian product** of n sets $A_1, A_2, \dots, A_n$ to be

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, \dots, a_n) : a_i \in A_i \text{ for } i = 1, \dots, n\}.$$

If $A_1 = A_2 = \cdots = A_n = A$, we often write A^n for $A_1 \times A_2 \times \cdots \times A_n$ (where A^n would be written n times). For example, the set $\mathbb{R}^3$ consists of all of 3-tuples of real numbers.

Subsets of $A \times B$ called relations. We will define a **mapping** or **function** f from a set A to a set B as a special type of relation $f \subseteq A \times B$ from a set A to a set B such that for every $a \in A$, there is a unique element $b \in B$ such that $(a, b) \in f$. Another way of saying this is that for every element a in A , f assigns a unique element in B . We usually write $f(a) = b$ or $a \mapsto b$ instead of writing down ordered pairs (a, b) . So if $f \subseteq A \times B$, we write

$$f(A) = \{f(a) : a \in A\} \subseteq B$$

called the **range** or **image** of f . We can think of the elements in the function's domain as input values and the elements in the function's range as output values.

where

$$\begin{aligned} F_{11}(u_n, v_n, \delta) &= \frac{(1+b)x^*\delta}{2a} - \frac{a \sin x^* u_n^2}{2} - \frac{(1+b)x^*\delta^2}{4a^2} \\ &\quad - \frac{a \cos x^* u_n^3}{6} - \frac{(1+b)x^*\delta^3}{8a^3} + O(\|(u_n, v_n, \delta)\|^4), \\ F_{12}(u_n, v_n, \delta) &= - \frac{(-x^*b^2 + 2a \sin x^* - 2x^*b - x^*) \delta}{2a} \\ &\quad + \frac{a(b-1) \sin x^* u_n^2}{2} + \frac{(1+b)^2 x^* \delta^2}{4a^2} - \cos x^* \delta u_n \\ &\quad + \frac{a(b-1) \cos x^* u_n^3}{6} + \frac{(1+b)^2 x^* \delta^3}{8a^3} + \frac{\sin x^* \delta u_n^2}{2} \\ &\quad + O(\|(u_n, v_n, \delta)\|^4). \end{aligned}$$

Consider the following system

$$\begin{pmatrix} u_{n+1} \\ v_{n+1} \\ \delta_{n+1} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_n \\ v_n \\ \delta_n \end{pmatrix} + \begin{pmatrix} \hat{F}_{11}(u_n, v_n, \delta_n) \\ \hat{F}_{12}(u_n, v_n, \delta_n) \\ 0 \end{pmatrix}, \quad (15)$$

where

$$a_{11} = -1 + a \cos x^*, \quad a_{12} = 1,$$

then the system (15) becomes

$$\begin{pmatrix} x_{n+1} \\ \tilde{y}_{n+1} \\ \tilde{\delta}_{n+1} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -b + a \cos x^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x}_n \\ \tilde{y}_n \\ \tilde{\delta}_n \end{pmatrix} + \begin{pmatrix} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ \tilde{F}_{12}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) \\ 0 \end{pmatrix}, \quad (16)$$

where

$$\begin{aligned} \tilde{F}_{11}(\tilde{x}_n, \tilde{y}_n, \tilde{\delta}_n) &= - \frac{\tan x^* \sec x^* \tilde{x}_n^2}{2a} - \frac{\tan x^* \tilde{x}_n \tilde{y}_n}{b-1} + \frac{\sin x^* \tan x^* \tilde{\delta}_n \tilde{x}_n}{2 \cos x^* (a-2b-2)} \\ &\quad - \frac{a \sin x^* \tilde{y}_n^2}{2(b-1)^2} + \frac{a \sin^2 x^* \tilde{y}_n \tilde{\delta}_n}{2(b-1)(a \cos x^* - b - 1)} \\ &\quad - \frac{\tilde{\delta}_n^2}{8(a \cos x^* - b - 1)^2 a^2} \left((2x^*(b+1)^3 + a^3 \sin^3 x^* \right. \\ &\quad \left. - 4ax^*(b+1)^2 \cos x^* + 2a^2 x^*(b+1) \cos^2 x^*) \right) \\ &\quad + \frac{\sec^2 x^* \tilde{x}_n^3}{6a^2} + \frac{\sec x^* \tilde{x}_n^2 \tilde{y}_n}{2a(b-1)} - \frac{\tan x^* \tilde{x}_n^2 \tilde{\delta}_n}{4a(a \cos x^* - b - 1)} \end{aligned}$$



第十一届清疏大学生数学竞赛班讲义

非数学类 & 数学类共用

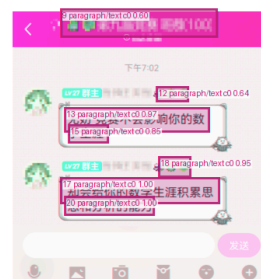
作者：清疏数学

组织：清疏大学

时间：July 7, 2024

版本：2.0

作者联系方式：QQ:937821418



本讲义将自动生成和 cctalk 用户绑定的 ID, 如果你的 ID 多次出现在公开环境, 我们会追究法律责任

仕.

Limits and Series (p.1)

1 paragraph/text c0 0.99

Problem Books in Mathematics

1 paragraph/text c0 0.99

Ovidiu Furdui

2 paragraph/text c0 1.00

3 paragraph/text c0 1.00

4 paragraph/text c0 1.00

5 paragraph/text c0 1.00

6 paragraph/text c0 1.00

7 paragraph/text c0 1.00

Limits, Series,
and Fractional
Part Integrals

8 paragraph/text c0 0.98

Problems in Mathematical Analysis



10 paragraph/text c0 0.94

Springer

block_layout_regression_v4

8.34

!

MATHCRAFT_FORCE_ORT_CPU MATHCRAFT_HOME

(永久生效):

输入高级环境变量

在 用户变量 中新建，变量名填入上面给出的名称，值填入 或路径

(仅当前终端):

在 中执行:

(永久生效):

在 或 末尾添加:

然后执行

(仅当前终端):

MATHCRAFT_FORCE_ORT_CPU 1
MATHCRAFT_HOME
HTTP_PROXY HTTPS_PROXY
MATHCRAFT_LOG_LEVEL
LATEXSNIPPER_FORCE_LINUX_GRAPHICS_FALLBACKS

LATEXSNIPPER_DISABLE_LINUX_GRAPHICS_FALLBACKS
LATEXSNIPPER_SHOW_CONSOLE